IF2240 Basdat

Reducing ER Diagrams to Relational Schemas

Type of Mapping from ER to Relational Model

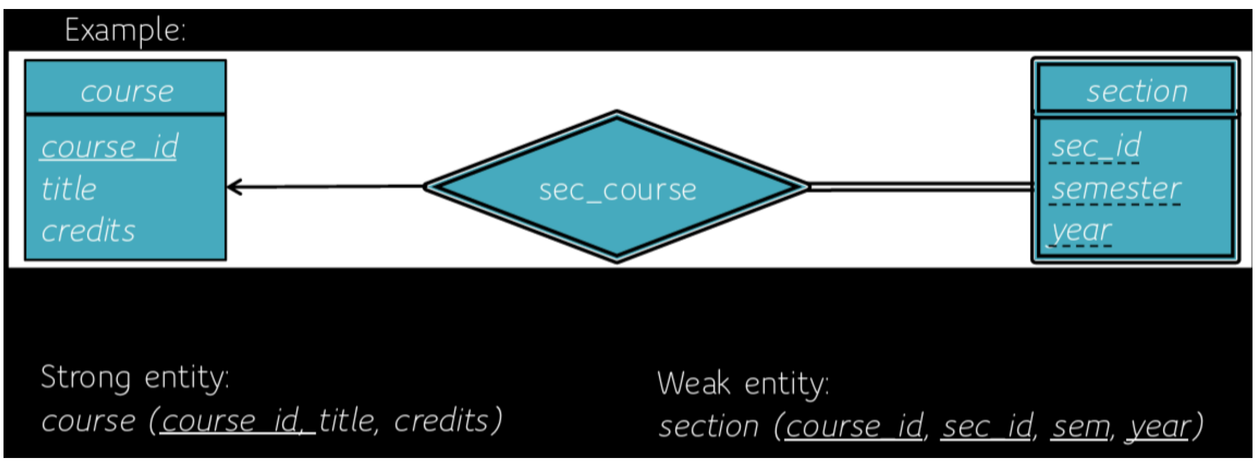
ER to Relational Mapping Algorithm

Mapping EER Model Constructs to Relations

Correspondence of Relational Model with ER Model

- Relations (tables) correspond with entity types and with many-to-many relationship types
- Columns correspond with attributes
- Rows correspond with entity instances and with many-to-many relationship instances

Representing Entity Sets



- A **strong** entity set reduces to a schema with the same attributes
- A **weak** entity set becomes a relation schema that includes a column for the primary key of the identifying strong entity se

Entity Sets with Composite Attributes

- Composite attributes are flattened out by creating a separate attribute for each component attribute
- Ignoring multivalued attributes and omitted derived attributes, extended instructor schema is:

instructor

ID
name
    first_name
    middle_initial
    last_name
address
    street
        street_number
        street_name
        apt_number
    city
    state
    zip
{ phone_number }
date_of_birth
age ( )

instructor(ID,
    first_name, middle_initial, last_name,
    street_number, street_name,
    apt_number, city, state, zip,
    date_of_birth)

Entity Sets with Multivalued Attributes

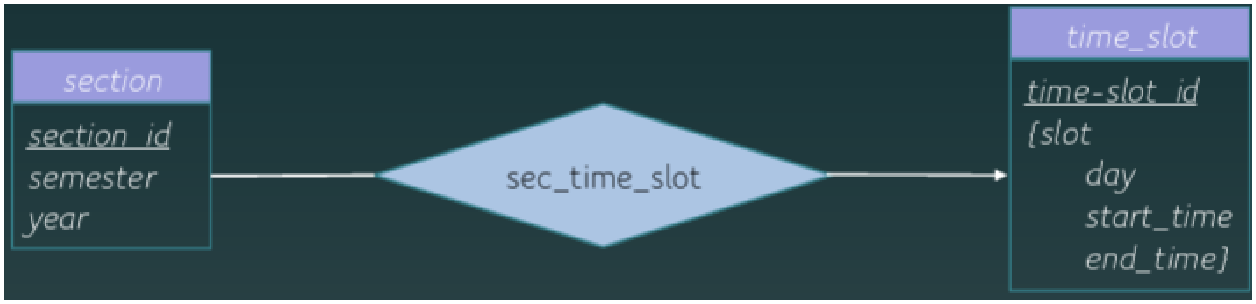
- A multivalued attribute M of an entity E is represented by a separate schema EM.
- Schema EM has attributes corresponding to the primary key of E and an attribute corresponding to multivalued attribute M.
- Take example form the instructor entity sets before.

Example: Multivalued attribute *phone_number* of *instructor* is represented by a schema:
    inst_phone (ID, phone_number)

Each value of the multivalued attribute maps to a separate tuple of the relation on schema *EM*
- For example, an *instructor* entity with primary key *22222* and phone numbers *456-7890* and *123-4567* maps to two tuples:
  (22222, 456-7890) and *(22222, 123-4567)*

→ tidak ada jaminan setiap ID memiliki phone number yang berbeda dan tidak ada jaminan sebuah phone number hanya dimiliki oleh sebuah ID

- Special case: ketika sebuah entity hanya memiliki primary key dan sebuah atribut multivalue:

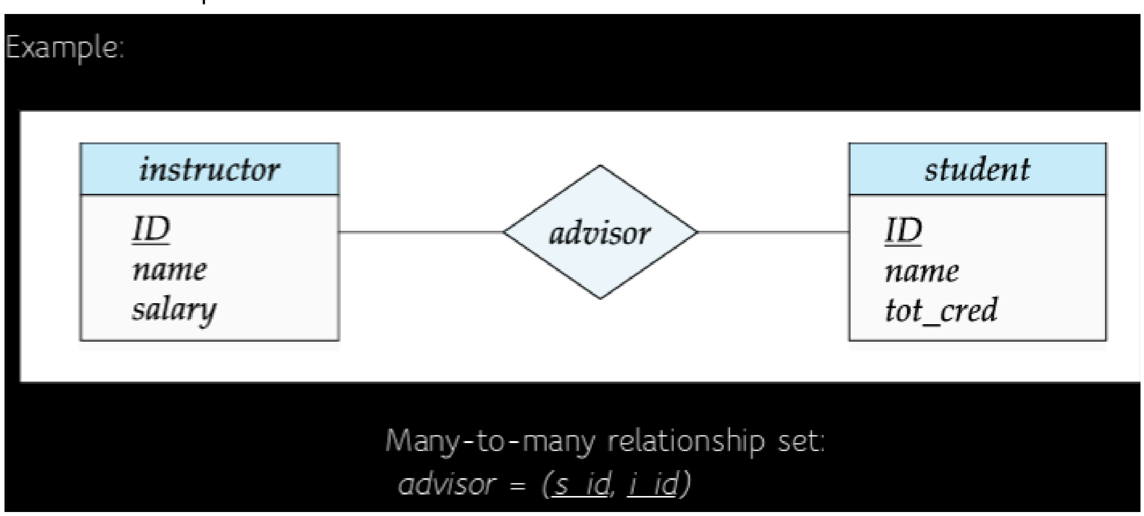


Derived Attributes

- Derived attributes are omitted, they are not implemented in the relational shema
- Derived attributes are presented using views

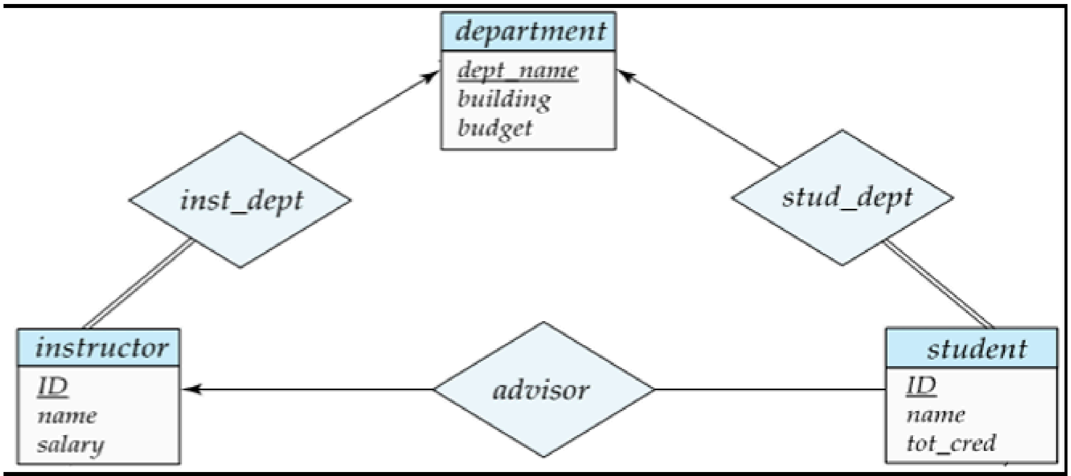
Representing Relationship Sets

- A many-to-many relationship set is represented as a schema with attributes for the primary keys of the two participating entity sets and any descriptive attributes of the relationship set.



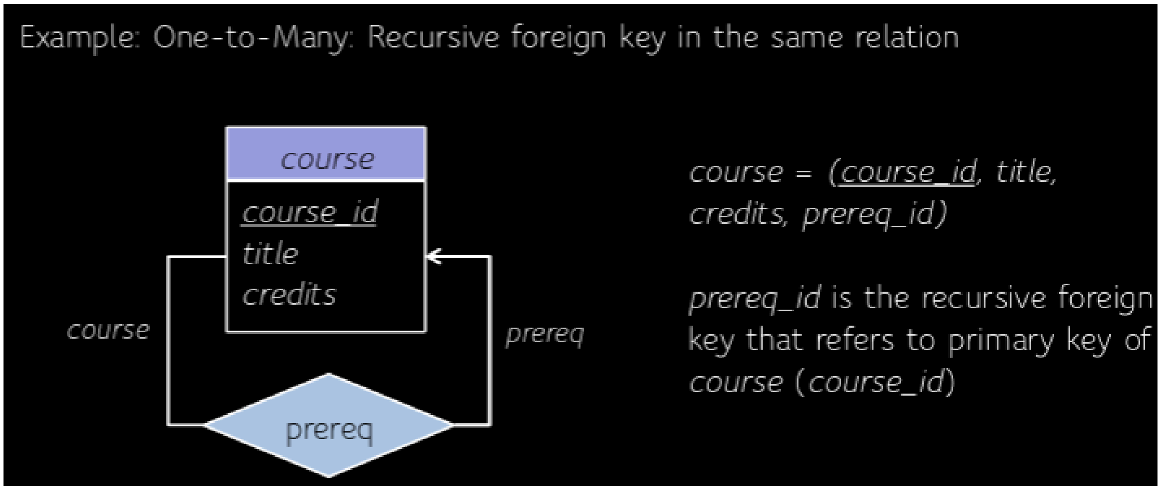
- many-to-one and one-to-many relationship sets that are total on the many-side can be represented by adding an extra attribute to the "many" side, containing the primary key of the "one" side.
- For one-to-one relationship sets, either side can be chosen to act as the "many" side. That is, an extra attribute can be added to either of the tables corresponding to the two entity sets

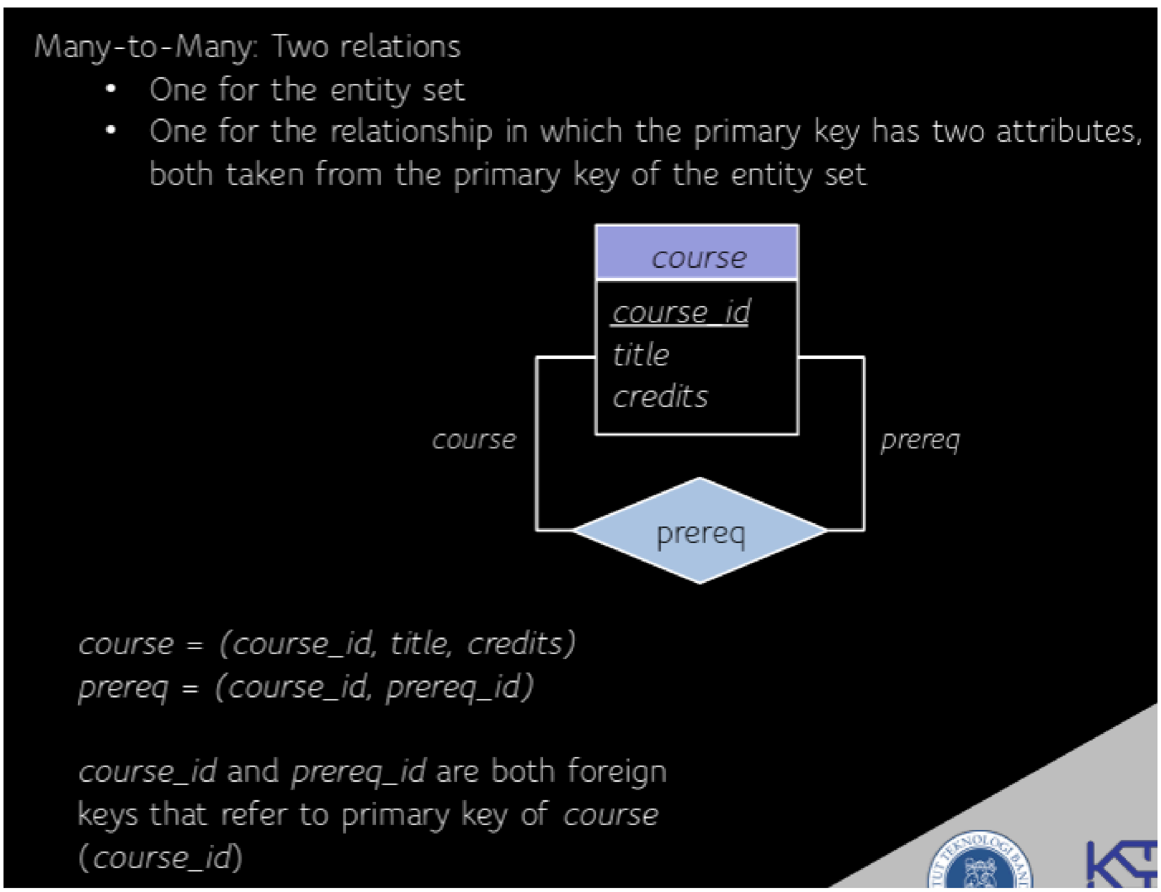
- If participation is partial on the “many” side, replacing a schema by an extra attribute in the schema corresponding to the “many” side could result in null value.
- Example:



Instead of creating a schema for relationship set inst_dept, add an attribute dept_name to the schema arising from entity set instructor.
instructor = (ID, name, salary, dept_name)
student = (ID, name, tot_credit, dept_name, instructorID)
department = (dept_name, building, budget)

Representing Unary Relationship





Representing Ternary and n-ary Relationship

Representing Specialization via Schemas

<table>
<tr><th>Method 1</th><th>Method 2</th></tr>
<tr><td>

Form a schema for the higher-level entity
Form a schema for each lower-level entity set, include primary key of higher-level entity set and local attributes
<table>
<tr><th>schema</th><th>attributes</th></tr>
<tr><td>person</td><td>ID, name, street, city</td></tr>
<tr><td>student</td><td>ID, tot_cred</td></tr>
<tr><td>employee</td><td>ID, salary</td></tr>
</table>

Drawback: getting information about

</td><td>

Form a schema for each entity set with all local and inherited attributes
<table>
<tr><th>schema</th><th>attributes</th></tr>
<tr><td>person</td><td>ID, name, street, city</td></tr>
<tr><td>student</td><td>ID, name, street, city, tot_cred</td></tr>
<tr><td>employee</td><td>ID, name, street, city, salary</td></tr>
</table>

If specialization is total, the schema for the generalized entity set not required to store information
Drawback: redundant

</td></tr>
</table>

Reducing Aggregation to Relational Schemas